Chapter 2 • Logarithms

Solved Exercise 2.2

1. Express each exponential statement in logarithmic form.

(i) $10^3 = 1000 \Leftrightarrow \log_{10}1000$
$= 3.$
(ii) $2^8 = 256 \Leftrightarrow \log_2256$
$= 8.$
(iii) $3^{-3} = \frac{1}{27} \Leftrightarrow \log_3(\frac{1}{27})$
$= -3.$
(iv) $20^2 = 400 \Leftrightarrow \log_{20}400$
$= 2.$
(v) $16^{-1/4} = \frac{1}{2} \Leftrightarrow \log_{16}(\frac{1}{2})$
$= \frac{-1}{4}$
(vi) $11^2 = 121 \Leftrightarrow \log_{11}121$
$= 2.$
(vii) $p = q^r \Leftrightarrow \log_q p$
$= r.$
(viii) $32^{-1/5} = \frac{1}{2} \Leftrightarrow \log_{32}(\frac{1}{2})$
$= \frac{-1}{5}$

2. Express each logarithmic statement in exponential form.

(i) $\log_5125 = 3 \Leftrightarrow 5^3$
$= 125.$
(ii) $\log_216 = 4 \Leftrightarrow 2^4$
$= 16.$
(iii) $\log_{23}1 = 0 \Leftrightarrow 23^0$
$= 1.$
(iv) $\log_55 = 1 \Leftrightarrow 5^1$
$= 5.$
(v) $\log_2(1/8) = -3 \Leftrightarrow 2^{-3}$
$= \frac{1}{8}$
(vi) $\frac{1}{2} = \log_93 \Leftrightarrow 9^{1/2}$
$= 3.$
(vii) $5 = \log_{10}100000 \Leftrightarrow 10^5$
$= 100,000.$
(viii) $\log_4(1/16) = -2 \Leftrightarrow 4^{-2}$
$= \frac{1}{16}$

3. Find x in each logarithmic equation.

(i) $\log_x64=3$
$\Rightarrow x^3=64=4^3$
$\Rightarrow x=4$ (base $x>0, x\neq1$).
4
(ii) $\log_51=x$
$\Rightarrow 5^x=1=5^0$
$\Rightarrow x=0.$
(iii) $\log_x8=1$
$\Rightarrow x^1=8$
$\Rightarrow x=8.$
(iv) $\log_{10}x=-3$
$\Rightarrow x=10^{-3}=\frac{1}{1000}$
(v) $\log_4x=\frac{3}{2}$
$\Rightarrow x=4^{3/2}=(\sqrt{4})^3=8.$
(vi) $\log_21024=x$
$\Rightarrow 2^x=1024=2^{10}$

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$\Rightarrow x=10.$

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